

Основы языка программирования Python

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Аннотации типов

```
def mul(a, b):  
    return a * b  
  
foo = mul(3.5, 2.0)  
bar = mul("spam", 3)  
  
print(f"{foo=}")  
print(f"{bar=}")
```

```
foo=7.0  
bar='spamspamspam'
```

```
def mul(a: float, b: float) -> float:  
    return a * b
```

```
foo = mul(3.5, 2.0)  
bar = mul("spam", 3)
```

```
print(f"{foo=}")  
print(f"{bar=}")
```

```
foo=7.0  
bar='spamspamspam'
```

Mypy



Mypy: Static Typing for Python

<https://mypy-lang.org>

<https://github.com/python/mypy>

Mypy is an optional static type checker for Python that aims to combine the benefits of dynamic (or "duck") typing and static typing. Mypy combines the expressive power and convenience of Python with a powerful type system and compile-time type checking. Mypy type checks standard Python programs; run them using any Python VM with basically no runtime overhead.

Пример использования Муру

```
> mypy example_typing_01.py
```

```
example_typing_01.py:5: error: Argument 1 to "mul" has incompatible  
type "str"; expected "float" [arg-type]  
Found 1 error in 1 file (checked 1 source file)
```

🔗 typing_01.py x |

```
1 def mul(a: float, b: float) -> float:  
2     return a * b
```

```
3  
4 foo = mul(3.5, 2.0)
```

```
5 bar = mul("spam", 3)
```

```
6  
7 print(f"{foo=}")  
8 print(f"{bar=}")
```

Argument of type "Literal['spam']" cannot be assigned to parameter "a" of type "float" in function "mul"

"Literal['spam']" is incompatible with "float" (Pyright reportArgumentType)

<https://github.com/microsoft/pyright/blob/main/docs/configuration.md#reportArgumentType>

```
def mul(a: str, b: int) -> str:  
    return a * b
```

```
bar = mul("spam", 3)
```

```
print(f"{bar=}")
```

```
# В качестве первого параметра у функции  
# ожидается строка или целое число.  
# Функция может вернуть строку или целое число  
def mul(a: str | int, b: int) -> str | int:  
    return a * b  
  
foo = mul(10, 3)  
bar = mul("spam", 3)
```



```
from typing import Union
```

```
# В качестве первого параметра у функции
```

```
# ожидается строка или целое число.
```

```
# Функция может вернуть строку или целое число
```

```
def mul(a: Union[str, int], b: int) -> Union[str, int]:  
    return a * b
```

```
foo = mul(10, 3)
```

```
bar = mul("spam", 3)
```

```
import math
```

```
def spam(a: float, b: float) -> None | float:  
    if a + b >= 0:  
        return math.sqrt(a + b)  
    return None
```

```
foo = spam(4, 5)  
bar = spam(-10, 1)
```

```
print(f"{foo=}")  
print(f"{bar=}")
```

```
foo=3.0  
bar=None
```

```
import math
from typing import Optional

def spam(a: float, b: float) -> Optional[float]:
    if a + b >= 0:
        return math.sqrt(a + b)
    return None

foo = spam(4, 5)
bar = spam(-10, 1)
```

```
def spam(foo: list[int]) -> dict[int, bool]:  
    return {n: n % 2 == 0 for n in foo}
```

```
foo = spam([10, 11, 12, 14, 17])  
print(f"{foo=}")
```

```
foo={10: True, 11: False, 12: True, 14: True, 17: False}
```

```
from typing import Dict, List
```

```
def spam(foo: List[int]) -> Dict[int, bool]:  
    return {n: n % 2 == 0 for n in foo}
```

```
foo = spam([10, 11, 12, 14, 17])  
print(f"{foo=}")
```

```
foo={10: True, 11: False, 12: True, 14: True, 17: False}
```

```
def spam(foo: list[int]) -> dict[int, bool]:  
    return {n: n % 2 == 0 for n in foo}
```

Результат вызова туру:

example_typing_09.py:4: error: Argument 1 to "spam"

has incompatible type "set[int]"; expected "list[int]" [arg-type]

Found 1 error in 1 file (checked 1 source file)

```
foo = spam({10, 11, 12, 14, 17})
```

```
print(f"{foo=}")
```

foo={17: False, 10: True, 11: False, 12: True, 14: True}

```
from collections.abc import Iterable
```

```
def spam(foo: Iterable[int]) -> dict[int, bool]:  
    return {n: n % 2 == 0 for n in foo}
```

```
foo = spam({10, 11, 12, 14, 17})  
print(f"{foo=}")
```

```
foo={17: False, 10: True, 11: False, 12: True, 14: True}
```